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Robust on-line partial discharge measurement of 35-kV cables in wind farms via dual sensors

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Abstract

Condition-based maintenance of cables in wind farms is critical for their operation reliability and stability. A promising solution to this task is to carry out on-line partial discharge measurement on the cables. However, impulse noise interferences from wind turbines or adjacent overhead lines could severely reduce the sensitivity of in field partial discharge measurement. Accordingly, this paper proposes a robust method to recognize the harmful partial discharge signal from the severe impulse noise. The method is based on joint analysis of the noisy partial discharge signals detected by two non-invasive partial discharge sensors, which are mounted on the double sides of the cable or on its terminal. The noiseless partial discharge signal and the impulse noise can be separated well via performing statistical clustering analysis on the pulse source location map or peak ratio vector obtained from the two detected noisy signal sequences. Finally, the effectiveness and feasibility of the proposed method is validated on two real partial discharge-affected 35-kV cables in an onshore wind farm.
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Keywords: Wind farms; Cables; Partial discharge; Impulse noise; Signal source separation

1. Introduction

As the increasing demand for renewable energy worldwide, a large number of wind farms have been built in recent years, and the number is expected to further increase in the future. The reliability of power components in wind farms has always been an issue, especially the premature failures of power cables [1]. These failures may be catastrophic and can lead to huge economic loss for the utilities. A promising solution to prevent the failures of cables is to conduct partial discharge (PD) diagnosis [2–5]. PD, a phenomenon of local dielectric breakdown, has proven to be a sensitive indicator of insulation deterioration of power cables and also the main reason of future failure [6]. According to the commissioning procedure of power cables, off-line PD diagnosis, which is conducted

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under an additional high voltage power supply (e.g., damped alternating current voltage [3]), is required before their full service to ensure their integrity [7,8]. However, PD activities in cables is a dynamic physical phenomenon related to operating voltage, environment and system transients [9], causing off-line PD diagnosis is not sufficient to predict possible failures during the full life cycle of the cables. Therefore, on-line PD diagnosis on cables is also required in order to support their efficient and reliable maintenance. However, it is yet challenging to achieve the acceptable sensitivity of on-line PD measurement of the cables, primarily due to various impulse noise interferences in wind farms. Since the cables are connected with many other power equipment, lots of impulse noise interferences (e.g., switching noise from gearbox in wind turbines and corona discharge from overhead lines) can couple to the cables, which may seriously reduce PD detection sensitivity. Fig. 1 collects a representative noisy PD signal that was measured by a high frequency current transformer mounted on a wind farm cable. It can be observed that the PD signal is almost drown out by the impulse noise.

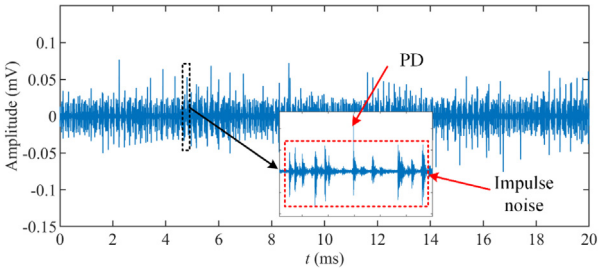


Fig. 1. Noisy partial discharge signal measured by a high frequency current transformer in a 35-kV cable in an onshore wind farm.

In recent years, many papers have been published on alternative impulse noise reduction approaches, mainly including statistical techniques, waveform analysis, and pattern recognition. In [10–12], statistical parameters, including mean, variance, kurtosis, etc., are used to distinguish the PD signal from the impulse noise. In [13,14], time and frequency-domain waveform parameters are defined to represent the differences between the PD signal and the impulse noise. However, once the impulse noise has waveform characteristics similar to PD pulses, the accuracy and reliability of these statistical techniques and waveform analysis methods are significantly reduced. It is important to point out that the above assumed situation may often occur in the cables in wind farms because the impulse noise here can be always complex and diverse, as shown in Fig. 1. In [15–18], pattern recognition methods, including genetic algorithm, support vector machine, artificial neural networks, and deep learning algorithm, are proposed to classify PD signals and noise signals including the impulse noise. However, a large number of samples are required to train the classifier, which is actually unrealistic for in field PD measurement.

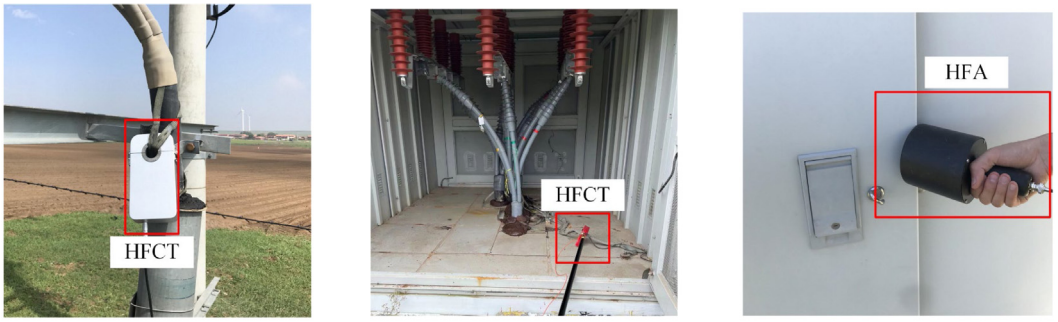
To overcome the above mentioned deficiencies, this paper proposes a robust method for identifying PD signals in wind farm cables. Since the waveform of the impulse noise is possibly similar to the PD signal, which leads to a difficulty to distinguish the PD signal from the impulse noise in the same signal sequence, we use two signal sequences detected by two sensors of different locations or types to inspect their difference. To identify a possible PD source in an underground cable, two high frequency current transformers are mounted on the double sides of the cable, and the PD pulses are extracted from the impulse noise according to the specific time difference between their arrival at the two sensors. In addition, to identify a possible PD source in a cable terminal, a high frequency current transformer and a high frequency antenna are mounted on the terminal, and PD pulses are extracted from the impulse noise according to the specific peak ratio of the two PD pulse sequences detected by the two sensors. Finally, the effectiveness and feasibility of the proposed method is validated on two real 35-kV cables with severe impulse noise disturbances.

2. Principle of the proposed method

In this section, non-invasive PD sensors for power cables in wind farms are first introduced, and then signal processing of the noisy signals detected by these sensors is described in details. Since the location of the underground cables is different from their terminals, PD detection of underground cables and their terminals is achieved via different approaches with different sensors, and thus discussed separately.

2.1. Non-invasive PD sensors

When PD activities occur in the middle of a cable, lots of high-frequency PD pulses are excited in the PD source and then propagate along the cable towards both ends. These PD pulses can arrive at the double-sided cable terminals and induce current pulses on the ground wire of the cable. Therefore, a high frequency current transformer (HFCT) can be mounted on the ground wire to detect the PD pulses. The HFCT sensor is non-invasive and can be easily mounted on running cables, thus allowing to easily carry out on-line PD measurement. In this work, two types of HFCT sensors are used: the one is the conventional HFCT sensor using coaxial signal cable transmission, as shown in Fig. 2(a), which is suitable for outdoor cable terminals; the other is the modified HFCT sensor using optical fiber signal transmission, as shown in Fig. 2(b), which is suitable for indoor cable terminals in branch boxes or switchgears. The optical fiber signal transmission can ensure safe and convenient installation of the HFCT sensor in the narrow space. The frequency pass-band of the two types of HFCT sensors is both from 0.5 to 50 MHz. The sensitivity of the conventional HFCT sensor is 5.6 mV/mA, which is higher than that of the modified HFCT sensor (3.3 mV/mA).



(a) (b) (c)

Fig. 2. Partial discharge (PD) measurement of cables in wind farms via (a) the high frequency current transformer (HFCT) with coaxial signal cable transmission, (b) the HFCT with optical fiber transmission, and (c) the high frequency antenna (HFA).

When PD activities occur in a cable terminal, the excited high-frequency PD pulses can not only induce current pulses on the ground wire, but also radiate electromagnetic waves in all directions from the PD source. Therefore, an additional high frequency antenna (HFA) sensor is used to detect the radiated electromagnetic waves in order to accurately identify the PD activities. For an indoor cable terminal, the HFA sensor is mounted close to the gap of the branch boxes or switchgears, as shown in Fig. 2(c); for outdoor terminals, the HFA sensor is close to it within a safe distance (see Fig. 7(a)). The frequency pass-band of the HFA sensor is in the range from 1 to 100 MHz. It is important to point out that the joint application of the two sensor types can separate the PD pulses occurring in the cable terminal from the impulse noise, which will be discussed in details in the following sub-sections.

2.2. Partial discharge detection for underground cables

In order to detect and locate the PD in the underground cable, two HFCT sensors are required to deploy on the double sides of the cable, as shown in Fig. 3. This allows us to obtain two PD signals on the double sides of the cable. Since source location of the PD is inherently different from the possible impulse noise source (the former locates in the middle of the cable, while the latter locates outside of the cable), separation of them is achieved by analyzing the time-of-arrival (TOA) difference between the two pulse sequences detected by the double-sided HFCT sensors.

Two discrete vectors $\boldsymbol{u}_{\mathrm{A}} = \{u_{\mathrm{A},1}, u_{\mathrm{A},2}, \ldots, u_{\mathrm{A},K}\}$ and $\boldsymbol{u}_{\mathrm{B}} = \{u_{\mathrm{B},1}, u_{\mathrm{B},2}, \ldots, u_{\mathrm{B},K}\}$ are defined to represent the noisy signals detected by the two HFCT sensors at Side A and Side B, respectively. It is important to point out that the two HFCT sensors should be triggered to collect data synchronously, which can be achieved by commercial Global Position Systems. Processing $\boldsymbol{u}_{\mathrm{A}}$ and $\boldsymbol{u}_{\mathrm{B}}$ to extract PD pulses from the impulse noise is completed via a three-step procedure as follows:

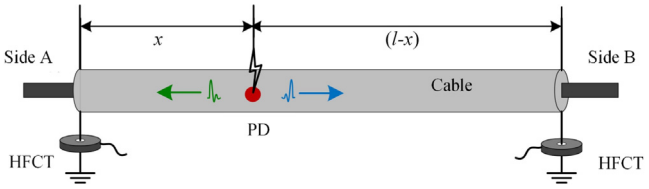


Fig. 3. Partial discharge (PD) detection of an underground cable via two high frequency current transformers (HFCTs).

Step (1) ***TOA extraction***: First, a time window, which is defined as N_{win} (e.g., 200), is moved on $\boldsymbol{u}_{\text{A}}$ and $\boldsymbol{u}_{\text{B}}$ to seek pulses. Second, taking a pulse identification in $\boldsymbol{u}_{\text{A}}$ as an example, once more than one of samples of $\boldsymbol{u}_{\text{A}}$ is greater than the set threshold in the time window $[N_{\text{start}}, N_{\text{start}} + N_{\text{win}}]$, N_{start} is defined as the start of a pulse. Then, the time window is sequentially moved until all samples of $\boldsymbol{u}_{\text{A}}$ in the time window $[N_{\text{end}}, N_{\text{end}} + N_{\text{win}}]$ are smaller than the threshold. N_{end} is defined as the end of the pulse. Therefore, the PD pulse can be extracted from $\boldsymbol{u}_{\text{A}}$ by selecting the samples in the time window of $[N_{\text{start}}, N_{\text{end}}]$. In addition, the above used threshold value is adaptively determined as 3σ, where σ is the standard derivation of the detected signal $\boldsymbol{u}_{\text{A}}$.

Due to attenuation and dispersion characteristics of the cable [19], pulses are distorted more or less on the cable, which could result in different time-to-peak values. Thus, onset detection is preferred because it is able to reduce the influence of the attenuation and dispersion. The energy criterion algorithm is selected due to its good accuracy and robustness to determine the onset point of a pulse even in high-level noise [20]. In the time window $[N_{\text{start}}, N_{\text{end}}]$, the pulse energy is defined as:

$$E = \sum_{k=N_{\text{start}}}^{N_{\text{end}}} u_{\text{A},k}^2 \tag{1}$$

which is computed to complete the EC-function as

$$EC(i) = \sum_{k=N_{\text{start}}}^{i} u_{\text{A},k}^2 - i\frac{E}{N_{\text{end}} - N_{\text{start}}}, i = [N_{\text{start}}, N_{\text{end}}] \tag{2}$$

The minimum value of $EC(i)$ represents the energy onset position and the TOA of the pulse. Applying Eqs. (1) and (2) to all pulses in $\boldsymbol{u}_{\text{A}}$ and $\boldsymbol{u}_{\text{B}}$ results into two TOA vectors $\boldsymbol{t}_{\text{A}} = \{t_{\text{A},1}, t_{\text{A},2}, \ldots, t_{\text{A},M}\}$ and $\boldsymbol{t}_{\text{B}} = \{t_{\text{B},1}, t_{\text{B},2}, \ldots, t_{\text{B},N}\}$, where N and M are the numbers of the pulses in $\boldsymbol{u}_{\text{A}}$ and $\boldsymbol{u}_{\text{B}}$, respectively. It is important to point out that extraction of PD pulses is equivalent to identify the elements in $\boldsymbol{t}_{\text{A}}$ and $\boldsymbol{t}_{\text{B}}$ that represent these PD pulses.

Step (2) ***Pulse source location mapping***: The location of PD source is inherently different from that of the impulse noise: the former is always in the middle of the cable, while the latter is out of the cable range. Pulse source location mapping is used to visualize the difference. A possible pulse source location x is calculated as:

$$x = \frac{l + (t_{A,i} - t_{B,j})v}{2} \quad s.t. |t_{A,i} - t_{B,j}| \leq \frac{l}{v}, i \in [1, M], j \in [1, N] \tag{3}$$

where l is the length of the cable, v is the propagation velocity of the pulse in the cable and depends on the materials of the cable (e.g., v can be approximately set to 172 m $(\mu\text{s})^{-1}$ for cross linked polyethylene cables), and the constraint condition of $|t_{\text{A},i} - t_{\text{B},j}| \leq l/v$ is used to limit the estimated pulse source location within the range of the cable. Applying Eq. (3) to all elements in $\boldsymbol{t}_{\text{A}}$ and $\boldsymbol{t}_{\text{B}}$ results in a pulse source location vector $\boldsymbol{x}$. Using the elements of $\boldsymbol{x}$ as the horizontal coordinate and the magnitudes of the corresponding pulses as vertical coordinate, a pulse source location map can be obtained. An example of the location map is shown in the panel(f) in Fig. 6. Since the impulse noise originates from the outside of the cable, pulses caused by the impulse noise locate on the double sides of the location map. In contrast, pulses caused by the PD activities in the cable locate in the middle of the location map. The PD pulses and impulse noise are clearly separated in the pulse source location map.

Step (3) ***Pulse source separation***: First, a distance window, which is defined as D_{win} (e.g., 10 m), is moved on $\boldsymbol{x}$ to seek pulse sources. Second, if the number of elements of $\boldsymbol{x}$ in the distance window $[D_{\text{start}}, D_{\text{start}} + D_{\text{win}}]$ is greater than the set threshold (e.g., 20), D_{start} is defined as the start of a pulse source. Then, the distance window is sequentially moved until the number of elements of $\boldsymbol{x}$ in the distance window $[D_{\text{end}}, D_{\text{end}} + D_{\text{win}}]$ are smaller than the threshold. D_{end} is defined as the end of the pulse source. Then, the pulses corresponding to the elements of $\boldsymbol{x}$ in the window $[D_{\text{start}}, D_{\text{end}}]$ are drawn in time-domain with their TOAs as the horizontal coordinate and their corresponding magnitudes as vertical coordinate. Repeating the above operation, multiple pulse sources can be extracted one by one, thus separating PD and impulse noise. Finally, the class of the PD pulses can be visually identified according to their uneven distribution characteristics in a power frequency cycle (e.g., 20 ms). This is because PD activities are associated with the power frequency voltage, and thus have typical power frequency phase distribution characteristics. In contrast, the impulse noise does not have the characteristics. Examples of the time-domain representation of the separated pulse sources are shown in the panel (g), (h), and (i) in Fig. 6.

2.3. Partial discharge detection for cable terminals

To detect possible PDs in a cable terminal, a HFCT sensor and a HFA sensor are required, as shown in Fig. 4. When a PD source occurs in a cable terminal joint, the generated PD signal can couple to the ground wire of the cable and also radiate in all directions. The HFCT sensor and HFA sensor both can detect different versions of the noisy PD signal. In addition to the PD signals, the HFCT sensor can also detect the impulse noise in the cable, and the HFA sensor can detect the other impulse noise in space. Since the parameters of the impulse noise detected by the HFCT sensor are different from that of the HFA sensor, joint analysis of the noisy signals collected by the two sensors can help to extract the useful PD signal.

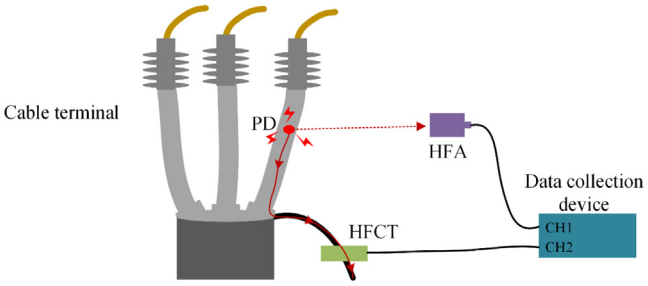


Fig. 4. Partial discharge (PD) detection of a cable terminal via joint application of a high frequency current transformer (HFCT) and a high frequency antenna (HFA).

Two discrete vectors $\boldsymbol{u}_{\text{C}} = \{u_{\text{C},1}, u_{\text{C},2}, \ldots, u_{\text{C},K}\}$ and $\boldsymbol{u}_{\text{D}} = \{u_{\text{D},1}, u_{\text{D},2}, \ldots, u_{\text{D,K}}\}$ are defined to represent the noisy signals detected by the HFCT sensor and HFA sensor, respectively. Processing the two vectors $\boldsymbol{u}_{\text{C}}$ and $\boldsymbol{u}_{\text{D}}$ to extract PD pulses from the impulse noise is completed via a three-step procedure as follows:

Step (1) ***Pulse peak extraction***: First, same as Step (1) in Section 2.2, the pulses in $\boldsymbol{u}_{\text{C}}$ and $\boldsymbol{u}_{\text{D}}$ are extracted and their TOA vectors $\boldsymbol{t}_{\text{C}} = \{t_{\text{C},1}, t_{\text{C},2}, \ldots, t_{\text{C,M}}\}$ and $\boldsymbol{t}_{\text{D}} = \{t_{\text{D},1}, t_{\text{D},2}, \ldots, t_{\text{D,N}}\}$ are calculated. Then, the peaks of the pulses are calculated as follows. Since the peak of a pulse always cannot be obtained by comparing the sample values, especially when the pulse and its reflections are superimposed on each other, we define the peak as the first point over the threshold where the trend of the first order difference changes from positive to negative or from negative to positive. Two peak vectors $\boldsymbol{p}_{\text{C}} = \{p_{\text{C},1}, p_{\text{C},2}, \ldots, p_{\text{C},M}\}$ and $\boldsymbol{p}_{\text{D}} = \{p_{\text{D},1}, p_{\text{D},2}, \ldots, p_{\text{D,N}}\}$ can be obtained via applying the above approach to all pulses in $\boldsymbol{u}_{\text{C}}$ and $\boldsymbol{u}_{\text{D}}$. It is important to point out that extraction of the PD pulses is equivalent to identify the elements in $\boldsymbol{p}_{\text{C}}$ and $\boldsymbol{p}_{\text{D}}$ that represent the PD pulses.

Step (2) ***Peak ratio histogram***: Peak ratio Q of the pulses in $\boldsymbol{p}_{\text{C}}$ and $\boldsymbol{p}_{\text{D}}$ is defined as:

$$Q = \log(\frac{p_{C,i}}{p_{D,j}}), i \in [1, M], j \in [1, N] \ s.t. |t_{C,i} - t_{D,j}| \leq 200 \text{ ns} \tag{4}$$

where the constraint condition of $|t_{C,i} - t_{D,j}| \leq 200$ ns is used to ensure that the pulses corresponding to $p_{C,i}$ and $p_{D,j}$ must originate from the same PD activity. Applying Eq. (4) to all elements in $\boldsymbol{t}_C$, $\boldsymbol{t}_D$, $\boldsymbol{p}_C$ and $\boldsymbol{p}_D$ results in a peak ratio vector $\boldsymbol{Q}$. Since the coupling paths from the PD source to the HFCT and HFA are fixed, the ratio between the two peak sub-sequences in $\boldsymbol{p}_C$ and $\boldsymbol{p}_D$ from the same PD source should be stabilized around a constant value. In contract, the impulse noise rarely satisfies this feature, or may has a different constant peak ratio value. Therefore, PD pulses and impulse noise can be separated via clustering the elements in $\boldsymbol{Q}$. The histogram of the peak ratio vector $\boldsymbol{Q}$ can be used to visualize the difference between the PD pulses and the impulse noise. An example of the histogram is shown in the panel (c) in Fig. 8.

Step (3) ***Pulse source separation***: Since the peak ratio concentration centers of possible classes of the elements in $\boldsymbol{Q}$ are unknown, which are different depending on the PD pulses and the impulse noise, we use K-means algorithm to automatically separate them. The K-means algorithm is a widely used method in signal processing, pattern recognition, and machine study fields, and has proven to be effective to cluster undefined data [21,22].

First, the number of class is defined as c referring to the histogram of $\boldsymbol{Q}$ Second, the function of K-means algorithm provided by MATLAB is used to cluster the elements in $\boldsymbol{Q}$ into c classes. Then, the pulses corresponding to the elements in each class are drawn in time-domain separately, with their TOAs as the horizontal coordinate and their corresponding magnitudes (e.g., the peak value in $\boldsymbol{p}_C$) as vertical coordinate. Finally, the class of the PD pulses is visually identify according to their power frequency phase distributions.

3. Validation experiment

3.1. Case I: Partial discharge in a cable middle joint

An on-site PD measurement was carried out on a 485 m 35 kV cable in an on-shore farm, as shown in Fig. 5(a). Two HFCT sensors were mounted on the double sides of the cable, respectively. The sampling ratio and time of the used data collection device were 125 MS/s and 20 ms, respectively. The detected signals and the results of signal processing are shown in Fig. 6. In panel (a) and (b), it can be observed that both of the detected signals include severe impulse noise, causing that it is very difficult to identify any PD pulse. In panel (c) and (d), TOAs and peaks of the pulses in the two detected signals are calculated. In panel (f), pulse location map is drawn, and it can be observed that three classes of pulses concentrate on different locations in the map: two classes locate at the double sides of the location mapping, whereas the other locate at the distance of 312 m from Side A, where there just is a middle joint. Finally, the time-domain representations of the three classes of pulses are drawn in panel (f), (g), and (h), respectively. It can be observed that the pulses in panel (g) have typical power frequency phase distribution characteristics, whereas the others do not. Combining the location information and power frequency phase distribution characteristics of the pulses in panel (g), it can be inferred that there should be a PD defect in the middle joint. After 5 days of the testing, the staff replaced the middle joint and dissected it, as shown in Fig. 5(b), where a visible damage of semiconducting layer caused by improper manual handling can be observed. Then we conducted a second PD test on the cable. No PD signals were found, thereby confirming that the damage of semiconducting layer had been the cause of the PD.

3.2. Case II: Partial discharge in a cable terminal joint

The other on-site PD measurement was carried out on a terminal joint of a 35-kV cable in the on-shore farm, as shown in Fig. 7(a). A HFCT sensor was mounted on the ground wire of the cable and a HFA sensor was held by hand close to the terminal. The sets of the used data collection devices were same to that in Case I. The detected signals and the result of signal processing are shown in Fig. 8. In panel (a) and (b), it can be observed that both of the detected signals include the severe impulse noise, causing that it is difficult to identify if there is any PD activity in the cable terminal. In panel (c), the histogram of the peak ratio vector of the two signals is calculated, and it can be observed that two classes of pulses concentrate on different locations in the histogram. The classification boundary between the two classes is calculated as 1.05 via the K-means clustering algorithm. Finally, the simplified time-domain representations of the two classes of pulses are shown in panel (d), and (e), respectively. It can be

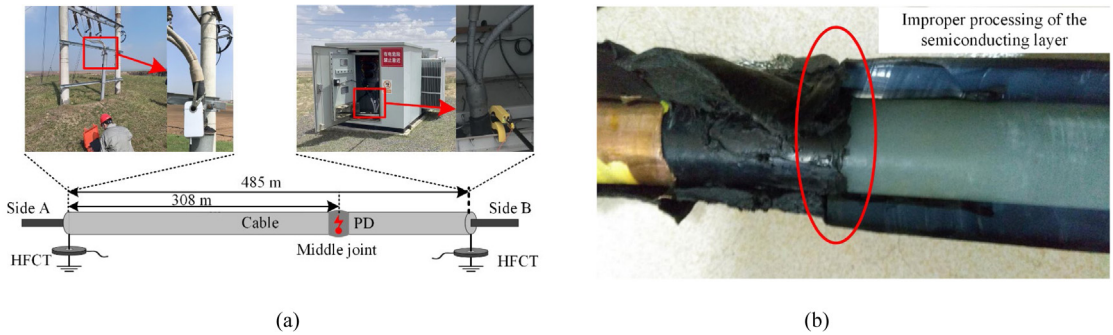


Fig. 5. Case I: layout of the double-sided high frequency current transformers (HFCTs) (a) and picture of the PD defect in the middle joint (b).

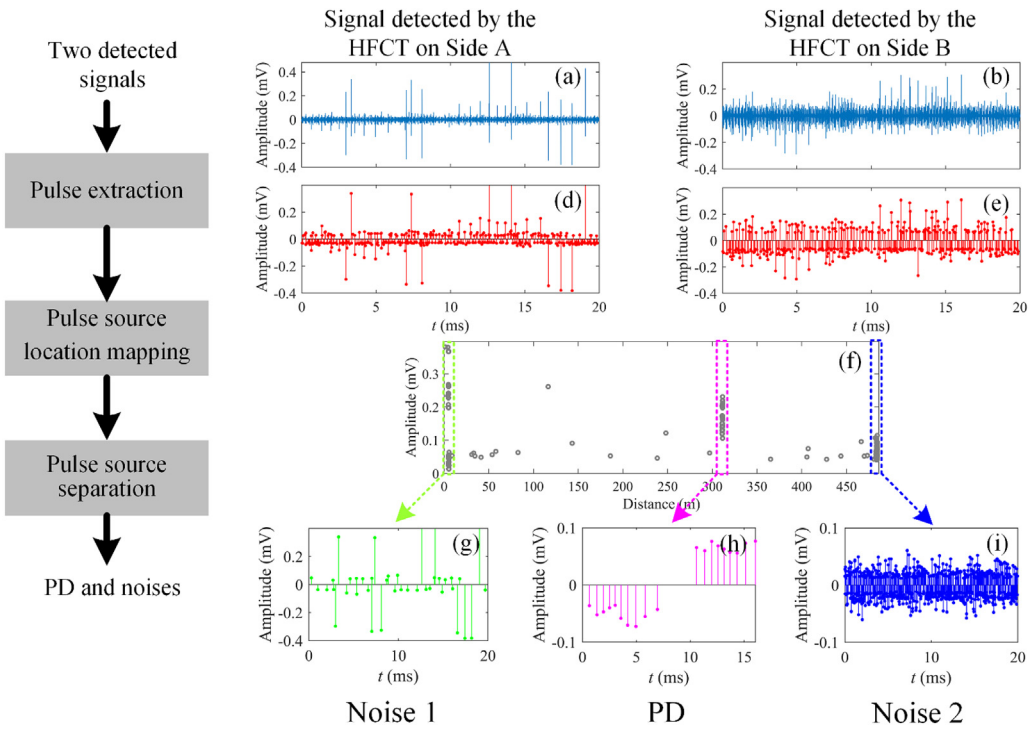


Fig. 6. Case I: signal processing of the signals detected by the double-sided high frequency current transformers (HFCTs).

observed that the pulses in panel (e) have typical power frequency phase distribution characteristics, whereas the other does not. Therefore, it can be inferred that there should be a PD defect in the cable terminal. In five days of the testing, the staff replaced the terminal joint and dissected it, as shown in Fig. 7(b), where several scratches caused by improper manual handling can be observed. Then we conducted a second PD test on the cable terminal. No PD signals were found, thereby confirming that the scratches had been the cause of the PD.

4. Conclusion

This paper highlights the severe impulse noise interference problem that may significantly reduce the sensitivity of on-line PD measurement of the cables in wind farms. To solve this problem, an alternative method based on joint application of two non-invasive PD sensors is proposed. The two sensors are deployed at the double sides of the underground cable or on its terminal, and can collect two versions of the noisy PD signal. This allows us to inspect



Fig. 7. Case II: layout of the high frequency current transformer (HFCT) and the high frequency antenna (HFA) for PD detection of an outdoor cable terminal joint.

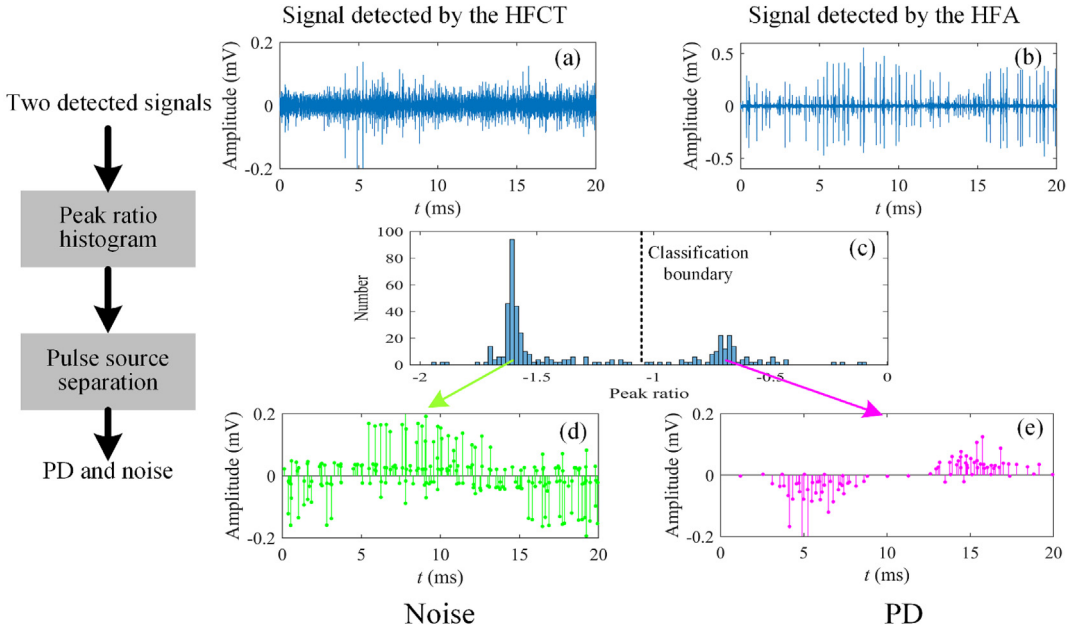


Fig. 8. Case II: signal processing of the signals detected by the high frequency current transformers (HFCT) and high frequency antenna (HFA).

the difference between the PD signal and the impulse noise according to some simple pulse parameters including time-of-arrival and peak value. Separation of the PD signal and the impulse noise are achieved by performing statistical analysis on the time-of-arrival difference or peak ratio between the two candidate pulse sequences collected by the two sensors respectively. Finally, the separated class of PD pulses is identified referring to their power frequency phase distribution characteristics. On-site PD measurement were carried out in two 35-kV cables in an on-shore wind farm, and the results reveal that the proposed method is effective to identify harmful PDs in cables and their accessories, in spite of the presence of very severe impulse noise.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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